

Tree Height Triangulation Shootout

Measure a tree without climbing it.

Win condition: Closest to instructor reference height.

THE SCIENCE YOU ARE USING

Angles give height. You can find a tree's height without touching the top. Sight the treetop through a paper clinometer to read an angle, measure your distance to the trunk, and the geometry of a right triangle gives the height. This is remote sensing with a protractor.

Accuracy from method. Small sighting errors grow over distance, so careful technique matters: level your eye, sight steadily, measure distance honestly. The team closest to the instructor's reference height wins.

HOW TO BUILD AND RUN IT

- 1 Build the clinometer.** Fold a paper clinometer with a weighted string plumb line to read angles.
- 2 Pace a known distance.** Measure your straight-line distance from the trunk and record it.
- 3 Sight the treetop.** Look along the clinometer at the very top and read the angle steadily.
- 4 Compute the height.** Use the angle, your distance, and your eye height to estimate the tree height.
- 5 Check against reference.** Compare to the instructor reference. The closest estimate wins.

Safety: Outdoor boundaries. Allergy: Pollen sensitivity.

MATERIALS

Paper clinometers	per team
String and washers	shared
Rulers and tape measures	shared
Clipboards	6

YOUR PLAN AND DATA

Clean, complete data is worth as much as a fast result.

Prediction (what will work best, and why):

The ONE variable we are testing:

Baseline result:

After redesign:

DEFEND YOUR DESIGN

What evidence made you change your design?

If you ran it again, what is the ONE thing you would change?

HOW YOU SCORE (100 POINTS)

Scoring criterion	Points
Accuracy	50
Method clarity	20
Speed	20
Teamwork	10
Total	100